c = width of class interval

x = mid-value of each class interval

where A is an assumed A.M.

Variance : Sum of the squares of the deviation from mean is known as Variance.

The square root of the variance is known as standard deviation.

NOTE

The simplified form of standard deviation formula may also be used

1. $\sigma = \frac{1}{n}\sqrt{n\sum d^2 - \left(\sum d\right)^2}$ (for raw data)
2. $\sigma = \frac{1}{N}\sqrt{N\sum fd^2 - \left(\sum fd\right)^2} \times C$

(for grouped data) where $d = (x\text{-A})/c$

Example 6.5

The following data gives the number of books taken in a school library in 7 days find the standard deviation of the books taken

7, 9, 12, 15, 5, 4, 11

Solution:

Actual mean method

$$\bar{x} = \frac{\sum x}{n}$$

$$\frac{7+9+.....+11}{7} = \frac{63}{7} = 9$$

x	$d = x - \bar{x}$	d^2
7	-2	4
9	0	0
12	3	9
15	6	36
5	-4	16
4	-5	25
11	2	4
		94

$$\sigma = \sqrt{\frac{\sum d^2}{n}}$$

$$= \sqrt{\frac{94}{7}}$$

$$= \sqrt{13.43}$$

$$= 3.66$$

NOTE

We can use two methods to find standard deviation

1. Direct method
2. Shortcut method

Ch6_11th Stats.indd 164 2/3/2019 11:01:57 AM